LESSON 11: SUBNETTING MASTERY

TRAINING FOCUS

This lesson is part of the FEMTECT Network Engineering Foundation course. It is written to help students build strong networking fundamentals through clear explanations, real enterprise context, practical troubleshooting logic, and certification-aware technical depth.

VISUAL DIAGRAM: SUBNET RANGE STRUCTURE

Visual flow

From	Relationship	To
192.168.10.0/26	connects to	Network Address; 192.168.10.0
192.168.10.0/26	connects to	Usable Hosts; 192.168.10.1 - 192.168.10.62
192.168.10.0/26	connects to	Broadcast Address; 192.168.10.63

This view helps students see that every IPv4 subnet has a beginning, a usable range, and an ending broadcast address.

DEFINITION AND TECHNICAL IN-DEPTH KNOWLEDGE

11.1 Subnetting

Subnetting is the process of dividing a larger IP network into smaller logical networks called subnets. A subnet is a smaller network created from a larger address block. Subnetting is one of the most important skills in networking because it allows administrators to organize IP addresses, control broadcast domains, improve security, support routing, and allocate addresses efficiently.

In Lesson 10, IPv4 addressing was introduced from first principles. Students learned that an IPv4 address has 32 bits, that a subnet mask separates the network portion from the host portion, and that a host uses its IP address and subnet mask to decide whether a destination is local or remote. Lesson 11 builds on that foundation and teaches how to calculate the actual network address, broadcast address, valid host range, CIDR prefix, block size, and variable-length subnet design.

Subnetting exists because real networks are not one giant flat network. An enterprise may need separate networks for staff, voice, guest Wi-Fi, servers, CCTV cameras, management, point-of-sale terminals, and cloud VPN connections. Each group may require a different number of IP addresses, different security controls, and different routing policies. Subnetting provides the addressing structure that makes this possible.

For CCNA and CompTIA Network+, subnetting is a high-value skill. It appears in addressing questions, routing questions, VLAN questions, DHCP design, access control lists, troubleshooting, and job interviews. A technician who can subnet confidently can understand network diagrams, identify configuration mistakes, and design practical IP addressing plans.

11.2 Technical and Business Importance of Subnetting

Subnetting matters because IP addressing must match the way the business network is organized. If one company places every device in one large IP network, broadcasts can become excessive, security boundaries are weak, and troubleshooting becomes harder. If the company divides the network into well-planned subnets, each department or function can have its own logical network.

Subnetting also matters because routers route between networks. A router does not simply route between individual devices; it uses network prefixes in its routing table. A well-designed subnet plan helps routers summarize routes, firewalls apply policies, DHCP servers hand out addresses correctly, and documentation

remain readable.

Subnetting saves address space. If a point-to-point WAN link needs only two usable IP addresses, assigning a full `/24` would waste 252 usable addresses. If a small printer VLAN needs 20 usable addresses, assigning a large block may waste space that another department needs. Subnetting allows the address space to match the requirement.

In troubleshooting, subnetting helps answer practical questions. Is this IP address in the same subnet as the gateway? What is the valid host range? Is the configured gateway actually usable in that subnet? Is the device using the network address by mistake? Is the device using the broadcast address by mistake? Is the subnet mask wrong?

11.3 SUBNETTING LANGUAGE

Before calculating, students must understand the vocabulary. Subnetting has several terms that appear repeatedly.

The network address identifies the subnet itself. It is not normally assigned to a host. For example, in `192.168.10.0/24`, the address `192.168.10.0` is the network address.

The broadcast address is the last address in the subnet. It is used to reach all hosts in that subnet and is not normally assigned to a host. For example, in `192.168.10.0/24`, the broadcast address is `192.168.10.255`.

The valid host range contains the usable addresses between the network address and the broadcast address. In `192.168.10.0/24`, the valid host range is `192.168.10.1` through `192.168.10.254`.

The subnet mask tells which bits identify the network and which bits identify hosts. CIDR notation writes this as a slash prefix such as `/24`, `/26`, or `/30`.

Subnetting Vocabulary Summary

Term	Meaning	Example in `192.168.10.0/24`
Network Address	Identifies the subnet	`192.168.10.0`
Broadcast Address	Last address in the subnet	`192.168.10.255`
First Usable Host	First assignable host address	`192.168.10.1`
Last Usable Host	Last assignable host address	`192.168.10.254`
Subnet Mask	Separates network and host bits	`255.255.255.0`
CIDR Prefix	Slash notation for network bits	`/24`

11.4 NETWORK, BROADCAST, AND HOST ADDRESSES

Every IPv4 subnet has a network address, broadcast address, and usable host range.

The network address is created when all host bits are 0. The broadcast address is created when all host bits are 1. Usable host addresses are the addresses between those two values.

Example:

Command / configuration block

```
Network:       192.168.10.0/24
Mask:          255.255.255.0
Network:       192.168.10.0
First Host:    192.168.10.1
Last Host:     192.168.10.254
Broadcast:     192.168.10.255
```

Why are the network and broadcast addresses not normally assigned to hosts? The network address represents the subnet itself. The broadcast address represents all hosts in the subnet. Assigning either one to a host would

create addressing confusion and communication problems.

In modern networks, some special point-to-point designs can use `/31` addressing, but for beginner CCNA and Network+ subnetting, students should first master the standard rule: usable hosts are total addresses minus 2.

Standard Usable Host Formula

Command / configuration block

```
Usable hosts = 2^host bits - 2
```

For a `/24`, there are 8 host bits.

Command / configuration block

```
2^8 = 256 total addresses
256 - 2 = 254 usable hosts
```

For a `/26`, there are 6 host bits.

Command / configuration block

```
2^6 = 64 total addresses
64 - 2 = 62 usable hosts
```

11.5 CIDR NOTATION

CIDR means Classless Inter-Domain Routing. CIDR notation writes the subnet mask as a slash followed by the number of network bits.

Example:

Command / configuration block

```
192.168.10.0/24
```

The `/24` means that 24 bits are network bits. Since IPv4 has 32 total bits, the remaining 8 bits are host bits.

Command / configuration block

```
32 total bits - 24 network bits = 8 host bits
```

CIDR replaced strict classful addressing because classful addressing wasted too much space. Instead of being limited to Class A, B, and C default masks, engineers can choose a prefix length that fits the network requirement.

Common CIDR Prefixes

CIDR	Subnet Mask	Host Bits	Total Addresses	Usable Hosts
/24	255.255.255.0	8	256	254
/25	255.255.255.128	7	128	126
/26	255.255.255.192	6	64	62
/27	255.255.255.224	5	32	30
/28	255.255.255.240	4	16	14
/29	255.255.255.248	3	8	6
/30	255.255.255.252	2	4	2

Students should memorize this table because it appears constantly in enterprise networking, CCNA questions, Network+ questions, firewall rules, VPN configuration, DHCP scopes, and cloud subnet design.

11.6 SUBNET MASK BINARY LOGIC

Subnet masks are easier to understand when viewed in binary. In a subnet mask, network bits are 1s and host bits are 0s.

Example `/24`:

Command / configuration block

```
Decimal Mask: 255.255.255.0
Binary Mask:  11111111.11111111.11111111.00000000
Prefix:       /24
```

Example `/26`:

Command / configuration block

```
Decimal Mask: 255.255.255.192
Binary Mask:  11111111.11111111.11111111.11000000
Prefix:       /26
```

The last octet `192` is `11000000` in binary. That means two bits in the fourth octet are borrowed for the network portion, leaving six host bits.

Common Last-Octet Masks

CIDR	Last Octet Binary	Last Octet Decimal	Block Size
/25	10000000	128	128
/26	11000000	192	64
/27	11100000	224	32
/28	11110000	240	16
/29	11111000	248	8
/30	11111100	252	4

The block size is the subnet increment. It tells where each subnet begins.

11.7 THE BLOCK-SIZE OR MAGIC-NUMBER METHOD

The block-size method is a fast way to calculate subnets. Many instructors call it the magic-number method.

The block size is found by subtracting the interesting octet value from 256.

Command / configuration block

```
Block size = 256 - interesting octet
```

The interesting octet is the octet in the subnet mask where the mask is not 255 and not 0.

Example:

Command / configuration block

```
Subnet Mask: 255.255.255.192
Interesting Octet: 192
Block Size: 256 - 192 = 64
```

This means the subnets increase by 64 in the fourth octet:

Command / configuration block

```
192.168.10.0
192.168.10.64
192.168.10.128
192.168.10.192
```

For each subnet, the broadcast address is one less than the next subnet. The usable range is between the network address and broadcast address.

Example `/26` Subnets

Subnet	Network Address	First Host	Last Host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

This method is fast, practical, and exam-friendly.

11.8 HOW TO FIND THE SUBNET OF A HOST ADDRESS

One of the most common tasks is identifying which subnet contains a host IP address.

Example:

Command / configuration block

```
Host IP: 192.168.10.77
Mask:    255.255.255.192
CIDR:    /26
```

Step 1: Identify the interesting octet.

The mask is `255.255.255.192`, so the interesting octet is the fourth octet.

Step 2: Calculate the block size.

Command / configuration block

```
256 - 192 = 64
```

Step 3: List the subnet increments.

Command / configuration block

```
0, 64, 128, 192
```

Step 4: Find where 77 fits.

77 is between 64 and 127, so the subnet is:

Command / configuration block

```
Network Address: 192.168.10.64
Broadcast:       192.168.10.127
Valid Hosts:     192.168.10.65 - 192.168.10.126
```

Therefore, `192.168.10.77/26` belongs to the `192.168.10.64/26` subnet.

Host Subnet Example

Item	Value
Host IP	`192.168.10.77`
Mask	`255.255.255.192`
CIDR	`/26`
Block Size	64
Network Address	`192.168.10.64`
First Host	`192.168.10.65`

Item	Value
Last Host	`192.168.10.126`
Broadcast	`192.168.10.127`

11.9 WORKED EXAMPLE: /27 SUBNETTING

Assume the company has this network:

Command / configuration block

```
192.168.20.0/24
```

The administrator wants smaller subnets using `/27`.

Step 1: Identify the mask.

Command / configuration block

```
/27 = 255.255.255.224
```

Step 2: Find the block size.

Command / configuration block

```
256 - 224 = 32
```

Step 3: List the subnet networks.

Command / configuration block

```
192.168.20.0
192.168.20.32
192.168.20.64
192.168.20.96
192.168.20.128
192.168.20.160
192.168.20.192
192.168.20.224
```

Step 4: Find the valid host ranges.

`/27` Subnet Table

Network	First Host	Last Host	Broadcast
192.168.20.0	192.168.20.1	192.168.20.30	192.168.20.31
192.168.20.32	192.168.20.33	192.168.20.62	192.168.20.63
192.168.20.64	192.168.20.65	192.168.20.94	192.168.20.95
192.168.20.96	192.168.20.97	192.168.20.126	192.168.20.127
192.168.20.128	192.168.20.129	192.168.20.158	192.168.20.159
192.168.20.160	192.168.20.161	192.168.20.190	192.168.20.191
192.168.20.192	192.168.20.193	192.168.20.222	192.168.20.223
192.168.20.224	192.168.20.225	192.168.20.254	192.168.20.255

Each `/27` subnet has 32 total addresses and 30 usable host addresses.

Command / configuration block

```
32 total - 2 = 30 usable
```

This size may be useful for small departments, printer networks, management networks, or small wireless segments.

11.10 WORKED EXAMPLE: /30 SUBNETTING

A `/30` subnet is commonly used in traditional point-to-point links because it provides exactly two usable host addresses.

Example network:

Command / configuration block

```
10.10.10.0/30
```

The `/30` mask is:

Command / configuration block

```
255.255.255.252
```

The block size is:

Command / configuration block

```
256 - 252 = 4
```

The subnets increase by 4:

Command / configuration block

```
10.10.10.0
10.10.10.4
10.10.10.8
10.10.10.12
```

First four `/30` subnets:

Network	First Host	Last Host	Broadcast
10.10.10.0	10.10.10.1	10.10.10.2	10.10.10.3
10.10.10.4	10.10.10.5	10.10.10.6	10.10.10.7
10.10.10.8	10.10.10.9	10.10.10.10	10.10.10.11
10.10.10.12	10.10.10.13	10.10.10.14	10.10.10.15

In a router-to-router link, one usable address is assigned to one router interface, and the second usable address is assigned to the other router interface.

Modern networks may use `/31` for point-to-point links to reduce waste, but `/30` remains common in training, exams, legacy networks, and many enterprise environments.

11.11 BORROWING BITS

Subnetting works by borrowing bits from the host portion and using them as additional network bits.

Start with a `/24`:

Command / configuration block

```
255.255.255.0
11111111.11111111.11111111.00000000
```

There are 8 host bits in the last octet.

If we borrow 2 bits, the prefix becomes `/26`:

Command / configuration block

```
255.255.255.192
11111111.11111111.11111111.11000000
```

Now there are 2 borrowed subnet bits and 6 remaining host bits.

Borrowing 2 bits creates:

Command / configuration block

```
2^2 = 4 subnets
```

Remaining 6 host bits provide:

Command / configuration block

```
2^6 - 2 = 62 usable hosts per subnet
```

Borrowing Bits from a /24

New Prefix	Bits Borrowed	Number of Subnets	Host Bits	Usable Hosts per Subnet
/25	1	2	7	126
/26	2	4	6	62
/27	3	8	5	30
/28	4	16	4	14
/29	5	32	3	6
/30	6	64	2	2

The more bits borrowed for subnets, the more subnets are created, but fewer host addresses remain in each subnet.

11.12 HOW TO CHOOSE A SUBNET SIZE

Choosing a subnet size means matching the number of usable host addresses to the business requirement.

If a department needs 50 devices, a `/26` provides 62 usable addresses and is usually enough. A `/27` provides only 30 usable addresses, which is too small. A `/25` provides 126 usable addresses, which works but may waste more addresses.

Subnet Size Selection

Requirement	Suitable Prefix	Usable Hosts	Comment
2 hosts	/30	2	Traditional point-to-point link
6 hosts	/29	6	Small device group
14 hosts	/28	14	Small management or printer subnet
30 hosts	/27	30	Small office or small VLAN
62 hosts	/26	62	Medium department
126 hosts	/25	126	Large department
254 hosts	/24	254	Common LAN subnet

In enterprise design, always allow reasonable growth. If a department has 28 current devices, a `/27` gives only 30 usable addresses and may be too tight once printers, phones, laptops, or temporary devices are added. A `/26` may be more practical.

However, do not waste address space carelessly. Good design balances current need, growth, summarization, documentation, and security boundaries.

11.13 VLSM FUNDAMENTALS

VLSM means Variable Length Subnet Mask. It allows different subnets inside the same larger address block to use different subnet masks.

VLSM exists because not every network segment needs the same number of IP addresses. A staff VLAN may need 100 usable addresses. A server subnet may need 30. A router-to-router link may need only 2. If every subnet used the same size, many addresses would be wasted.

With VLSM, the engineer starts with a larger block and allocates smaller blocks based on host requirements.

Example requirement:

Command / configuration block

```
Address Block: 192.168.50.0/24

Staff VLAN:       100 hosts
Voice VLAN:        50 hosts
Server VLAN:       25 hosts
Management VLAN:   10 hosts
WAN Link:           2 hosts
```

The correct VLSM method is to allocate from largest requirement to smallest requirement.

Why largest first? Larger subnets need bigger address blocks. If small subnets are assigned randomly first, the remaining space may become fragmented and unable to fit a larger subnet cleanly.

VLSM Planning Table

Requirement	Needed Hosts	Selected Prefix	Usable Hosts	Reason
Staff VLAN	100	/25	126	Smallest block that supports 100
Voice VLAN	50	/26	62	Smallest block that supports 50
Server VLAN	25	/27	30	Smallest block that supports 25
Management VLAN	10	/28	14	Smallest block that supports 10
WAN Link	2	/30	2	Traditional point-to-point link

VLSM Allocation Example

Network Use	Network Address	Prefix	Usable Host Range	Broadcast
Staff VLAN	192.168.50.0	/25	192.168.50.1 - 192.168.50.126	192.168.50.127
Voice VLAN	192.168.50.128	/26	192.168.50.129 - 192.168.50.190	192.168.50.191
Server VLAN	192.168.50.192	/27	192.168.50.193 - 192.168.50.222	192.168.50.223
Management VLAN	192.168.50.224	/28	192.168.50.225 - 192.168.50.238	192.168.50.239
WAN Link	192.168.50.240	/30	192.168.50.241 - 192.168.50.242	192.168.50.243

This leaves additional addresses from `192.168.50.244` through `192.168.50.255` for future small allocations, depending on design.

11.14 ENTERPRISE SUBNETTING EXAMPLE

Imagine a manufacturing company with one office and one shop floor. The company has staff computers, IP phones, CCTV cameras, production machines, guest Wi-Fi, and network management devices.

The company receives this private address block:

Command / configuration block

```
10.20.0.0/22
```

The engineer may divide it into subnets like this:

Function	Subnet	Purpose
Staff	10.20.0.0/24	Office computers and laptops
Voice	10.20.1.0/25	IP phones
CCTV	10.20.1.128/26	Cameras and NVR devices
Shop Floor	10.20.2.0/24	Production devices and controllers
Management	10.20.3.0/27	Switches, APs, firewalls, UPS management
Guest Wi-Fi	10.20.3.64/26	Internet-only guest users

This design is better than placing every device in one subnet because each function can have its own DHCP scope, firewall policy, monitoring baseline, and troubleshooting boundary.

For example, guest Wi-Fi should not have direct access to production machines. CCTV cameras should not be mixed with staff laptops. Management interfaces should be protected from normal user networks. Subnetting supports these boundaries.

11.15 COMMON SUBNETTING MISTAKES

Subnetting mistakes are common, especially when students calculate too quickly or skip the block-size logic.

One common mistake is assigning the network address to a host. For example, assigning `192.168.10.64/26` to a PC is wrong because `.64` is the network address for that subnet.

Another mistake is assigning the broadcast address to a host. In the same `/26` subnet, `192.168.10.127` is the broadcast address and should not be assigned to a host.

Another common mistake is choosing a subnet that is too small. If a department needs 40 hosts, a `/27` is not enough because it provides only 30 usable host addresses. A `/26` is needed.

A serious troubleshooting mistake is using the wrong subnet mask. Two devices may appear to have similar IP addresses but disagree about whether they are local. For example, one device may use `/24` while another uses `/26`. This can cause intermittent or confusing communication problems.

Common Mistakes Table

Mistake	Example	Result
Assigning network address	`192.168.10.64/26` on a host	Host address invalid
Assigning broadcast address	`192.168.10.127/26` on a host	Host address invalid
Wrong mask	One host `/24`, another `/26`	Local/remote decision mismatch
Too-small subnet	`/27` for 40 hosts	Not enough addresses
Overlapping subnets	`192.168.10.0/25` and `192.168.10.64/26`	Routing and policy confusion
Ignoring growth	Exactly 30 hosts in `/27`	No room for expansion

11.16 TROUBLESHOOTING SUBNETTING PROBLEMS

Subnetting troubleshooting requires careful calculation. A technician should not only look at the IP address. The IP address, subnet mask, default gateway, and intended subnet must be checked together.

Troubleshooting steps:

1. Identify the host IP address.

2. Identify the subnet mask or CIDR prefix.

3. Calculate the network address.

4. Calculate the broadcast address.

5. Identify the valid host range.

6. Confirm the default gateway is inside the same subnet.

7. Check whether another device is using the same IP address.

8. Confirm the VLAN and DHCP scope match the subnet.

9. Test local gateway reachability.

10. Test remote reachability after local subnet correctness is confirmed.

Example:

Command / configuration block

```
Host IP:         192.168.30.70
Subnet Mask:     255.255.255.192
Default Gateway: 192.168.30.1
```

The mask is `/26`. The block size is 64. The `/26` subnets are:

Command / configuration block

```
192.168.30.0
192.168.30.64
192.168.30.128
192.168.30.192
```

The host `192.168.30.70` belongs to:

Command / configuration block

```
Network:    192.168.30.64
Broadcast:  192.168.30.127
Host Range: 192.168.30.65 - 192.168.30.126
```

The configured gateway `192.168.30.1` is not in the same subnet. The correct gateway should be an address inside `192.168.30.65 - 192.168.30.126`, commonly something like `192.168.30.65` or `192.168.30.126`, depending on the design.

Subnetting Troubleshooting Table

Symptom	Possible Subnetting Cause	What to Calculate
Host cannot ping gateway	Gateway outside host subnet	Host range and gateway address
Some devices communicate, others fail	Inconsistent subnet masks	Network and broadcast per device
DHCP clients get wrong network	Wrong DHCP scope or VLAN	VLAN-to-subnet mapping
Routing table has confusing paths	Overlapping subnets	Prefix ranges and route specificity
Host has address ending in subnet boundary	Network or broadcast assigned	Block size and valid host range

11.17 WINDOWS, LINUX, AND CISCO IOS PRACTICAL COMMANDS

Windows Commands

Command / configuration block

```
ipconfig
ipconfig /all
ping <default-gateway>
arp -a
tracert <destination>
```

On Windows, `ipconfig /all` is the fastest way to check the IP address, subnet mask, default gateway, DHCP status, and DNS servers. If the subnet mask is wrong, the device may make the wrong local-versus-remote decision.

Linux Commands

Command / configuration block

```
ip addr
ip route
ip neigh
ping <default-gateway>
traceroute <destination>
```

On Linux, `ip addr` shows the prefix length directly, such as `192.168.10.25/24`. `ip route` shows connected networks and the default route. If the default gateway is missing or outside the connected subnet, remote connectivity will fail.

Cisco IOS Commands

Command / configuration block

```
show ip interface brief
show running-config interface gigabitEthernet 0/0
show ip route connected
show arp
ping <destination>
traceroute <destination>
```

On Cisco routers and Layer 3 switches, `show ip interface brief` quickly confirms interface addressing and status. `show ip route connected` shows networks that are directly connected based on interface IP addresses and masks.

11.19 CYBERSECURITY RELEVANCE

Subnetting is a security skill because security policy is often written around IP networks. Firewalls, ACLs, VPNs, monitoring tools, SIEM platforms, and endpoint management systems all depend on accurate network ranges.

If subnets are poorly designed, sensitive devices may share a network with ordinary users or guests. This increases risk. For example, CCTV cameras, payment systems, server management interfaces, and industrial control systems should not usually share the same subnet as guest Wi-Fi users.

Subnetting also helps incident response. If security logs show suspicious traffic from `10.20.3.77`, a documented subnet plan should quickly tell the analyst whether that address belongs to guest Wi-Fi, staff, servers, cameras, or management.

Incorrect subnetting can create security gaps. An ACL that permits `192.168.10.0/24` may allow more devices than intended if the real requirement was only `192.168.10.64/26`. A firewall rule with the wrong prefix can expose too much or block legitimate traffic.

11.20 CLOUD NETWORKING RELEVANCE

Cloud networking uses subnetting heavily. When creating a cloud virtual network or VPC, the engineer chooses a larger CIDR block and then divides it into subnets for public-facing resources, private application servers, databases, load balancers, VPN gateways, and management services.

Example:

Command / configuration block

```
Cloud VPC: 10.100.0.0/16
Public Subnet A:  10.100.1.0/24
Private App A:    10.100.10.0/24
Database A:       10.100.20.0/24
VPN Subnet:       10.100.250.0/24
```

Cloud subnets must not overlap with on-premises networks if VPN or private connectivity is required. If the office uses `10.100.0.0/16` and the cloud also uses `10.100.0.0/16`, routing across the VPN becomes confusing or impossible without redesign or translation.

VLSM thinking helps cloud engineers allocate space cleanly and leave room for future regions, availability zones, and environments such as development, testing, and production.

12. KEY TERMS / GLOSSARY

Term	Meaning
Subnetting	Dividing a larger IP network into smaller logical networks.
Subnet	A smaller network created from a larger address block.
Network Address	Address that identifies the subnet itself.
Broadcast Address	Last address in a subnet, used to reach all hosts in that subnet.
Valid Host Range	Usable addresses between the network and broadcast addresses.
CIDR Notation	Slash-prefix notation showing how many bits are network bits.
Prefix Length	Number of network bits in CIDR notation.
Host Bits	Bits left for host addressing after the network bits.
Block Size	Subnet increment used to identify subnet boundaries.
Magic Number	Another name for the block size in fast subnetting.
Interesting Octet	The subnet mask octet where subnet increments occur.
VLSM	Variable Length Subnet Mask, using different mask sizes in one design.
Overlapping Subnets	Subnets whose address ranges intersect, causing routing or policy confusion.
Usable Hosts	Assignable host addresses in a subnet, normally total addresses minus 2.
Point-to-Point Link	Link connecting two network devices, often using a small subnet such as `/30`.

13. COMPARISON TABLES

13.1 CIDR, Mask, and Host Capacity

CIDR	Subnet Mask	Block Size	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

13.2 Fixed-Length Subnetting vs VLSM

Feature	Fixed-Length Subnetting	VLSM
Mask Sizes	Same mask for all subnets	Different masks based on need
Efficiency	May waste addresses	More efficient
Simplicity	Easier to calculate	Requires careful planning
Enterprise Use	Useful for simple designs	Common in real designs
Risk	Oversized or undersized subnets	Overlap if poorly planned

13.3 Network vs Broadcast vs Host Address

Address Type	Host Bits	Assign to Host?	Purpose
Network Address	All 0s	No	Identifies subnet
Usable Host	Mixed values	Yes	Assigned to devices
Broadcast Address	All 1s	No	Reaches all hosts in subnet

13.4 Subnetting Design Choices

Design Choice	Advantage	Risk
Larger subnet	More growth room	More broadcasts and wasted addresses
Smaller subnet	Efficient address use	May run out of addresses
VLSM	Best fit per requirement	Requires careful documentation
Route summarization	Smaller routing tables	Requires planned contiguous blocks

14. REAL-WORLD EXAMPLE / ANALOGY

Subnetting is like dividing a large building into departments. The building has one overall address, but inside it there may be separate areas for finance, engineering, reception, security, and storage. Each department has its own room range and access rules.

In networking, the larger IP block is like the building. Each subnet is like a department area. The network address marks the beginning of the area. The broadcast address marks the end. The valid host range is the usable space inside.

This analogy helps, but the technical skill is calculation. A network engineer must be able to calculate where each subnet starts, where it ends, and which addresses can safely be assigned.

15. QUICK-RECAP CHEAT SHEET

- Subnetting divides a larger network into smaller networks.
- A subnet has a network address, valid host range, and broadcast address.
- The network address has all host bits set to 0.
- The broadcast address has all host bits set to 1.
- Usable hosts are normally `2^host bits - 2`.
- CIDR notation shows the number of network bits.
- `/24` equals `255.255.255.0`.
- `/25` equals `255.255.255.128` and supports 126 usable hosts.
- `/26` equals `255.255.255.192` and supports 62 usable hosts.
- `/27` equals `255.255.255.224` and supports 30 usable hosts.
- `/28` equals `255.255.255.240` and supports 14 usable hosts.
- `/29` equals `255.255.255.248` and supports 6 usable hosts.
- `/30` equals `255.255.255.252` and supports 2 usable hosts.
- Block size equals `256 - interesting octet`.
- The broadcast address is one less than the next subnet.
- First usable host is network address plus 1.
- Last usable host is broadcast address minus 1.
- VLSM uses different subnet masks inside the same larger address block.
- In VLSM, allocate from largest host requirement to smallest.
- The default gateway must be inside the host's subnet.
- Overlapping subnets cause routing and policy confusion.

7. LESSON ASSESSMENT

This assessment is intentionally focused. Students should answer clearly, use correct networking terminology, and show practical troubleshooting thinking.

7.1 Theory Questions

1. Explain the main networking concept covered in this lesson and describe its role in a real enterprise network.

2. Describe two technical problems that can occur when this lesson's concept is poorly designed or incorrectly configured.

3. Explain how a network technician or junior network engineer would verify that this part of the network is working correctly.

4. Describe one security, reliability, or performance concern related to this topic.

5. Explain how this topic connects to CCNA, Network+, IT support, or junior network engineering work.

7.2 Multiple-Choice Questions

1. What is the best reason for studying this lesson's topic?

A. It helps students understand and support real network operation. B. It replaces the need for troubleshooting. C. It removes the need for IP addressing. D. It is only useful for memorizing definitions.

2. When troubleshooting a problem related to this lesson, what should be done first?

A. Gather evidence and identify the exact scope of the problem. B. Replace every network device immediately. C. Delete the running configuration without backup. D. Assume the Internet service provider is always responsible.

3. Which habit best reflects professional network engineering practice?

A. Verify changes with commands, tests, and documented results. B. Make random changes until something works. C. Ignore diagrams and rely only on memory. D. Avoid documenting successful fixes.

4. Which item is most useful when proving that a network feature is working?

A. A successful test result compared with the expected result. B. A guess from another user. C. A device name with no configuration evidence. D. A screenshot unrelated to the problem.

5. What should students connect this lesson to during hands-on practice?

A. Addressing, device roles, commands, verification, and troubleshooting. B. Only typing speed. C. Only cable color. D. Only memorizing vendor logos.

6. In an enterprise environment, why is documentation important for this topic?

A. It helps future troubleshooting, audits, handover, and change control. B. It makes testing unnecessary. C. It disables security risks automatically. D. It replaces the need for correct configuration.

7. Which approach gives the strongest learning outcome?

A. Combining technical explanation with lab practice and verification. B. Reading commands without understanding them. C. Skipping troubleshooting steps. D. Memorizing answers without using the network.

8. If a configuration change causes a new outage, what should the technician use?

A. A rollback plan and saved baseline information. B. A random new subnet. C. A different monitor. D. A guess without evidence.

9. Which result shows that a student understands the lesson at a practical level?

A. The student can explain the concept, configure or identify it, test it, and troubleshoot failure. B. The student can only repeat the title. C. The student avoids using commands. D. The student ignores enterprise examples.

10. Why are scenario questions included in this course?

A. They train students to apply concepts to real network problems. B. They remove the need for labs. C. They focus only on spelling. D. They are unrelated to IT jobs.

7.3 Scenario-Based Questions

1. A user or department reports a connectivity problem related to this lesson's topic. Write the first five checks you would perform, the commands or tools you would use, and the evidence you expect to collect. 2. You are asked to explain this topic to a new IT support intern before a hands-on lab. Write a short technical explanation that connects the concept to real devices, real traffic flow, and troubleshooting.

7.4 Answer Guide

Theory answers should be marked for technical accuracy, clear explanation, correct terminology, and practical relevance to enterprise networking. Scenario answers should show a logical troubleshooting sequence, evidence gathering, likely causes, verification steps, and a professional final recommendation.

Multiple-choice answers: 1. A; 2. A; 3. A; 4. A; 5. A; 6. A; 7. A; 8. A; 9. A; 10. A